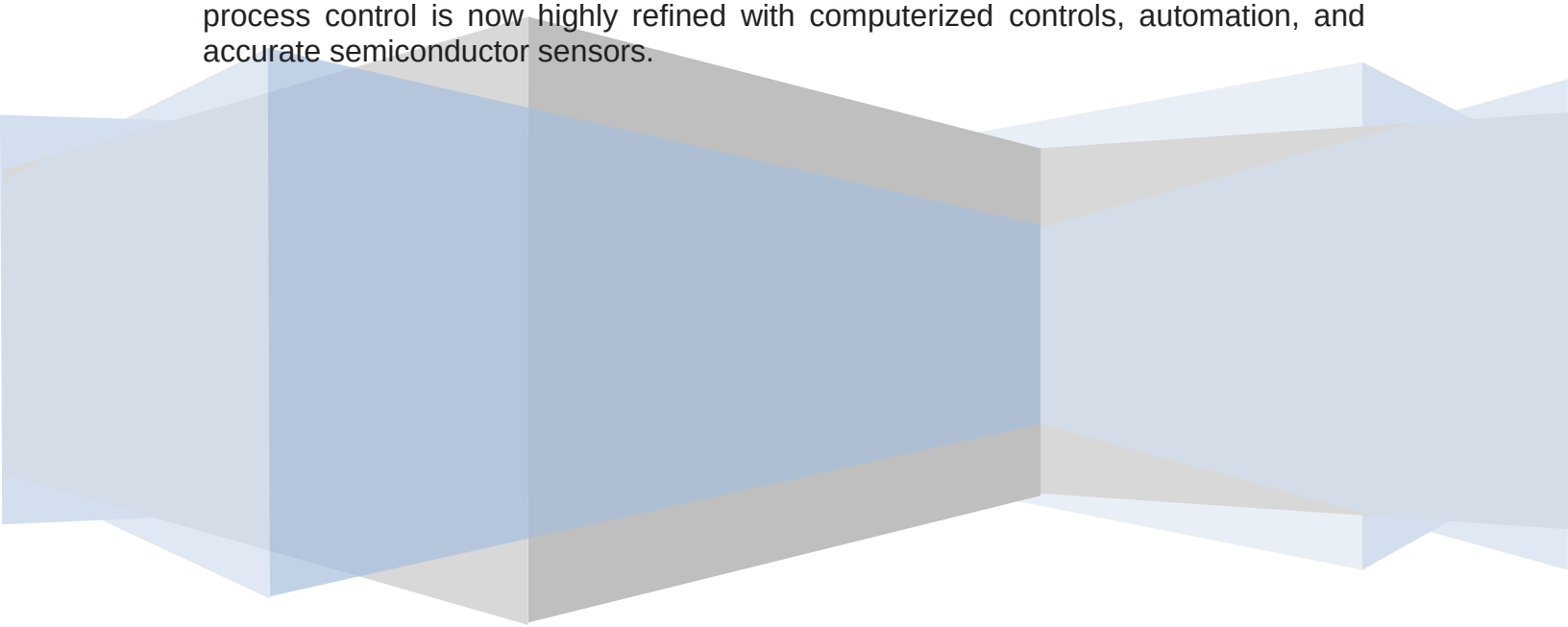
CHAPTER 1

INTRODUCTION TO PROCESS INSTRUMENTATION

The technology of controlling a series of events to transform a material into a desired end product is called process control. For instance, the making of fire could be considered a primitive form of process control. Industrial process control was originally performed manually by operators. Their sensors were their sense of sight, feel, and sound, making the process totally operator-dependent. To maintain a process within broadly set limits, the operator would adjust a simple control device.

Instrumentation and control slowly evolved over the years, as industry found a need for better, more accurate, and more consistent measurements for tighter process control. The first real push to develop new instruments and control systems came with the Industrial Revolution, and World Wars I and II added further to the impetus of process control.

Feedback control first appeared in 1774 with the development of the fly-ball governor for steam engine control, and the concept of proportional, derivative, and integral control during World War I. World War II saw the start of the revolution in the electronics industry, which has just about revolutionized everything else. Industrial process control is now highly refined with computerized controls, automation, and accurate semiconductor sensors.



1.2 Process Control

Process control can take two forms:

i) ***Sequential control***, which is an event-based process in which one event follows another until a process sequence is complete.

ii) ***Continuous control***, which requires continuous monitoring and adjustment of the process variables. However, continuous process control comes in many forms, such as domestic water heaters and heating, ventilation, and air conditioning (HVAC), where the variable temperature is not required to be measured with great precision, and complex industrial process control applications, such as in the petroleum or chemical industry, where many variables have to be measured simultaneously with great precision. These variables can vary from temperature, flow, level, and pressure, to time and distance, all of which can be interdependent variables in a single process requiring complex microprocessor systems for total control. Due to the rapid advances in technology, instruments in use today may be obsolete tomorrow. New and more efficient measurement techniques are constantly being introduced. These changes are being driven by the need for higher accuracy, quality, precision, and performance. Techniques that were thought to be impossible a few years ago have been developed to measure parameters.

1.2.1 Sequential Process Control

Control systems can be sequential in nature, or can use continuous measurement; both systems normally use a form of feedback for control. Sequential control is an event-based process, in which the completion of one event follows the completion of another, until a process is complete, as by the sensing devices. Figure 1.1 shows an example of a process using a sequencer for mixing liquids in a set ratio. The sequence of events is as follows:

1. Open valve A to fill tank A.
2. When tank A is full, a feedback signal from the level sensor tells the sequencer to turn valve A Off.
3. Open valve B to fill tank B.
4. When tank B is full, a feedback signal from the level sensor tells the sequencer to turn valve B Off.
5. When valves A and B are closed, valves C and D are opened to let measured quantities of liquids A and B into mixing tank C.
6. When tanks A and B are empty, valves C and D are turned off.
7. After C and D are closed, start mixing motor, run for set period.
8. Turn Off mixing motor.
9. Open valve F to use mixture.

10. The sequence can then be repeated after tank C is empty and Valve F is turned Off.

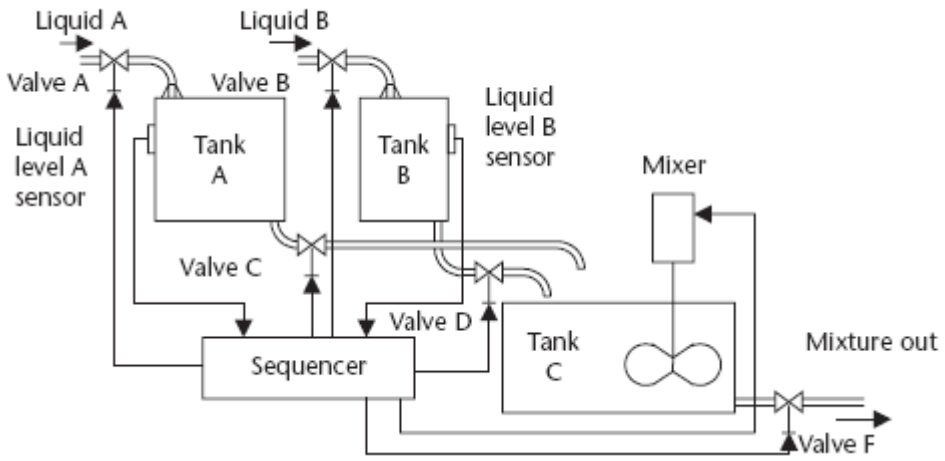


Figure 1.1:

1.2.2 Continuous Process Control

Continuous process control falls into two categories:

- Elementary On/Off action,
- Continuous control action.

On/Off action is used in applications where the system has high inertia, which prevents the system from rapid cycling. This type of control only has only two states, On and Off; hence, its name. This type of control has been in use for many decades, long before the introduction of the computer. HVAC is a prime example of this type of application. Such applications do not require accurate instrumentation. In HVAC, the temperature (measured variable) is continuously monitored, typically using a bimetallic strip in older systems and semiconductor elements in newer systems, as the sensor turns the power (manipulated variable) On and Off at preset temperature levels to the heating/cooling section.

Continuous process action is used to continuously control a physical output parameter of a material. The parameter is measured with the instrumentation or sensor, and compared to a set value. Any deviation between the two causes an error signal to be generated, which is used to adjust an input parameter to the process to correct for the output change. An example of an unsophisticated automated control process is shown in Figure 1.2. A float in a swimming pool is used to continuously monitor the level of the water, and to bring the water level up to a set reference point

when the water level is low. The float senses the level, and feedback to the control valve is via the float arm and pivot. The valve then controls the flow of water (manipulated variable) into the swimming pool, as the float moves up and down.

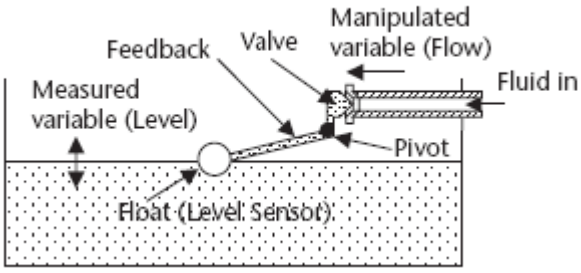


Figure 1.2:

A more complex continuous process control system is shown in Figure 1.3, where a mixture of two liquids is required. The flow rate of liquid A is measured with a differential pressure (Dp) sensor, and the amplitude of the signal from the Dp measuring the flow rate of the liquid is used by the controller as a reference signal (set point) to control the flow rate of liquid B. The controller uses a Dp to measure the flow rate of liquid B, and compares its amplitude to the signal from the Dp monitoring the flow of liquid A. The difference between the two signals (error signal) is used to control the valve, so that the flow rate of liquid B (manipulated variable) is directly proportional to that of liquid A, and then the two liquids are combined.

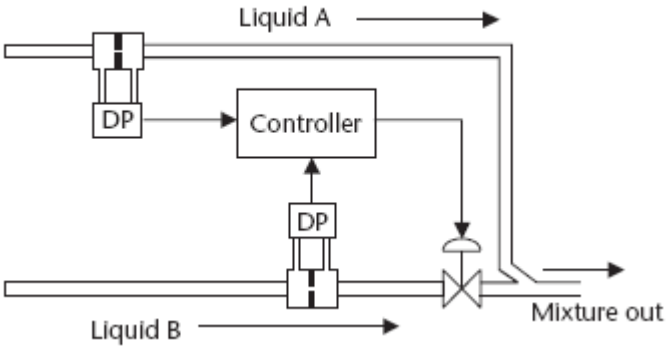


Figure 1.3:

1.3 Definition of the Elements in a Control Loop

In any process, there are a number of inputs (i.e., from chemicals to solid goods). These are manipulated in the process, and a new chemical or component emerges at the output. To get a more comprehensive look at a typical process control system, it will be broken down into its various elements. Figure 1.4 is a block diagram of the elements in a continuous control process with a feedback loop. Process is a sequence of events designed to control the flow of materials through a number of steps in a plant to produce a final utilitarian product or material. The process can be a simple process with few steps, or a complex sequence of events with a large number of interrelated variables. The examples shown are single steps that may occur in a process.

Measurement is the determination of the physical amplitude of a parameter of a material; the measurement value must be consistent and repeatable. Sensors are typically used for the measurement of physical parameters. A sensor is a device that can convert the physical parameter repeatedly and reliably into a form that can be used or understood. Examples include converting temperature, pressure, force, or flow into an electrical signal, measurable motion, or a gauge reading. In Figure 1.3, the sensor for measuring flow rates is a Dp cell.

Error Detection is the determination of the difference between the amplitude of the measured variable and a desired set reference point. Any difference between the two is an error signal, which is amplified and conditioned to drive a control element. The controller sometimes performs the detection, while the reference point is normally stored in the memory of the controller.

Controller is a microprocessor-based system that can determine the next step to be taken in a sequential process, or evaluate the error signal in continuous process control to determine what action is to be taken. The controller can normally condition the signal, such as correcting the signal for temperature effects or nonlinearity in the sensor. The controller also has the parameters of the process input control element, and conditions the error sign to drive the final element. The controller can monitor several input signals that are sometimes interrelated, and can drive several control elements simultaneously. The controllers are normally referred to as programmable logic controllers (PLC). These devices use ladder networks for programming the control functions.

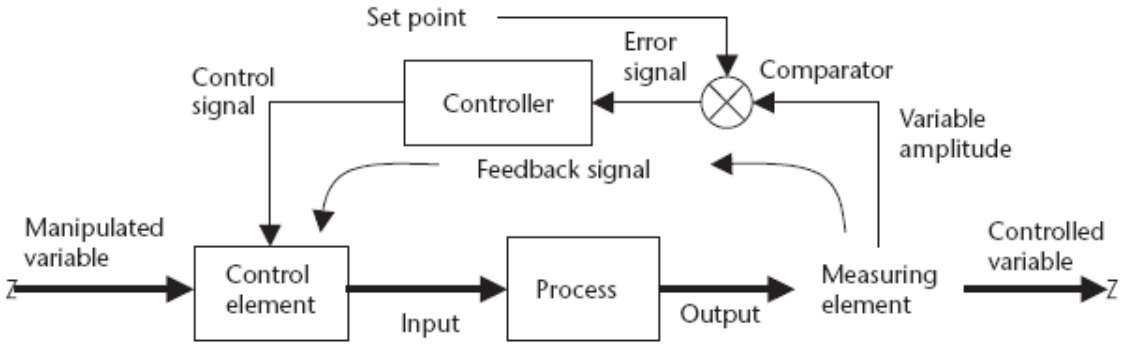


Figure 1.4:

Control Element is the device that controls the incoming material to the process (e.g., the valve in Figure 1.3). The element is typically a flow control element, and can have an On/Off characteristic or can provide liner control with drive. The control element is used to adjust the input to the process, bringing the output variable to the value of the set point.

The control and measuring elements in the diagram in Figure 1.4 are oversimplified, and are broken down in Figure 1.5. The **measuring element** consists of a sensor to measure the physical property of a variable, a **transducer** to convert the sensor signal into an electrical signal, and a **transmitter** to amplify the electrical signal, so that it can be transmitted without loss.

The **control element** has an actuator, which changes the electrical signal from the controller into a signal to operate the valve, and a control valve. In the feedback loop, the controller has memory and a summing circuit to compare the set point to the sensed signal, so that it can generate an **error signal**. The controller then uses the error signal to generate a correction signal to control the valve via the actuator and the input variable. The function and operation of the blocks in different types of applications will be discussed in a later chapter. The definitions of the terms used are given at the end of the chapter.

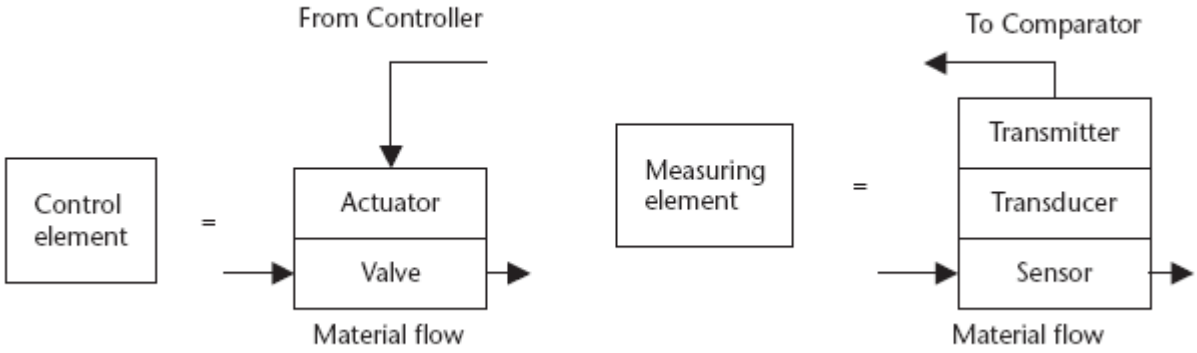


Figure 1.5:

1.4 Instrumentation and Sensors

The operator's control function has been replaced by instruments and sensors that give very accurate measurements and indications, making the control function totally operator-independent. The processes can be fully automated. Instrumentation and sensors are an integral part of process control, and the quality of process control is only as good as its measurement system. The subtle difference between an instrument and a sensor is that an **instrument** is a device that measures and displays the magnitude of a physical variable, whereas a **sensor** is a device that measures the amplitude of a physical variable, but does not give a direct indication of the value. The same physical parameters normally can be applied to both devices

1.4.1 Instrument Parameters

The choice of a measurement device is difficult without a good understanding of the process. All of the possible devices should be carefully considered. It is also important to understand instrument terminology.

Process Instrumentation Terminology gives the definitions of the terms used in instrumentation in the process control sector. Some of the more common terms are discussed below.

Accuracy of an instrument or device is the error or the difference between the indicated value and the actual value. Accuracy is determined by comparing an indicated reading to that of a known standard. Standards can be calibrated devices, and may be obtained from the National Institute of Standards and Technology (NIST). The NIST is a government agency that is responsible for setting and maintaining standards, and developing new standards as new technology requires it. Accuracy depends on linearity, hysteresis, offset, drift, and sensitivity. The resulting discrepancy is stated as a plus-or-minus deviation from true, and is normally specified as a percentage of reading, span, or of full-scale reading or deflection (% FSD), and can be expressed as an absolute value. In a system where more than one deviation is involved, the total accuracy of the system is statistically the root mean square (rms) of the accuracy of each element.

Exercise 1.1
A pressure sensor has a span of 25 to 150 psi. Specify the error when measuring 107 psi, if the accuracy of the gauge is (a) ±1.5% of span, (b) ±2% FSD, and (c) ±1.3% of reading.

Exercise 1.2
A pressure sensor has an accuracy of ±2.2% of reading, and a transfer function of 27 mV/kPa. If the output of the sensor is 231 mV, then what is the range of pressures that could give this reading?

Linearity is a measure of the proportionality between the actual values of a variable being measured and the output of the instrument over its operating range. The deviation from true for an instrument may be caused by one or several of the above factors affecting accuracy, and can determine the choice of instrument for a particular application. Figure 1.6 shows a linearity curve for a flow sensor, which is the output from the sensor versus the actual flow rate. The curve is compared to a best-fit straight line. The deviation from the ideal is 4 cm/min., which gives a linearity of ±4% of FSD.

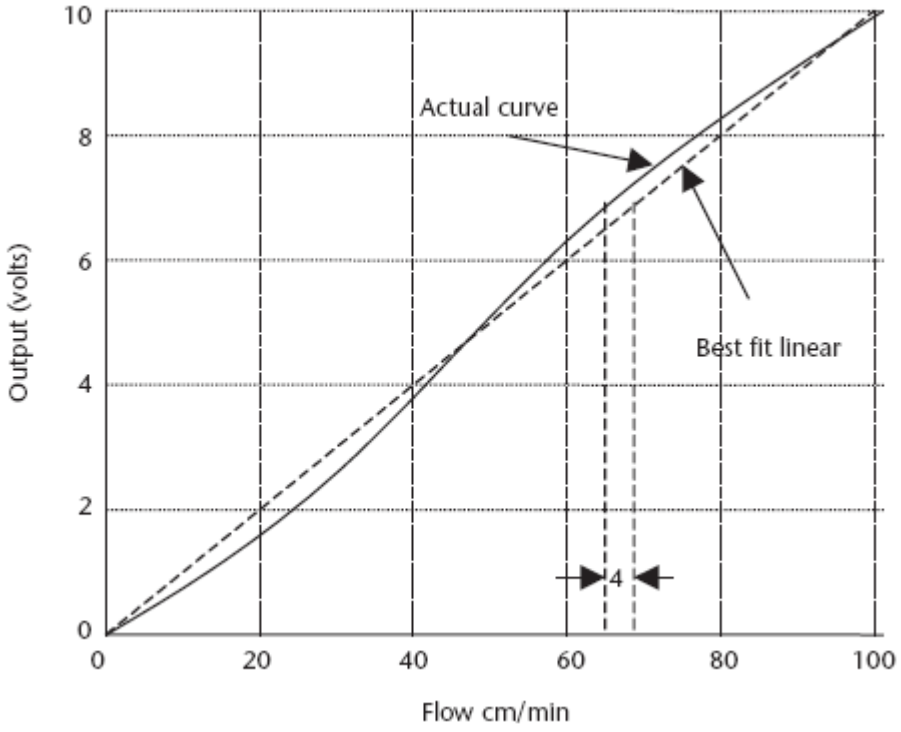


Figure 1.6:

Sensitivity is a measure of the change in the output of an instrument for a change in the measured variable, and is known as a *transfer function*. For example, when the output of a flow transducer changes by 4.7 mV for a change in flow of 1.3 cm/s, the sensitivity is 3.6 mV/cm/s. High sensitivity

in an instrument is desired, since this gives a higher output, but has to be weighed against linearity, range, and accuracy.

Reproducibility is the inability of an instrument to consistently reproduce the same reading of a fixed value over time under identical conditions, creating an uncertainty in the reading.

Resolution is the smallest change in a variable to which the instrument will respond. A good example is in digital instruments, where the resolution is the value of the least significant bit.

Exercise 1.3
A digital meter has 10-bit accuracy. What is the resolution on the 16V range?

Hysteresis is the difference in readings obtained when an instrument approaches a signal from opposite directions. For example, if an instrument reads a midscale value beginning at zero, it can give a different reading than if it read the value after making a full-scale reading. This is due to stresses induced into the material of the instrument by changing its shape in going from zero to full-scale deflection. A hysteresis curve for a flow sensor is shown in Figure 1.7, where the output initiating from a zero reading and initiating from a maximum reading are different. For instance, the output from zero for a 50 cm/min is 4.2V, compared to 5.6V when reading the same flow rate after a maximum reading.

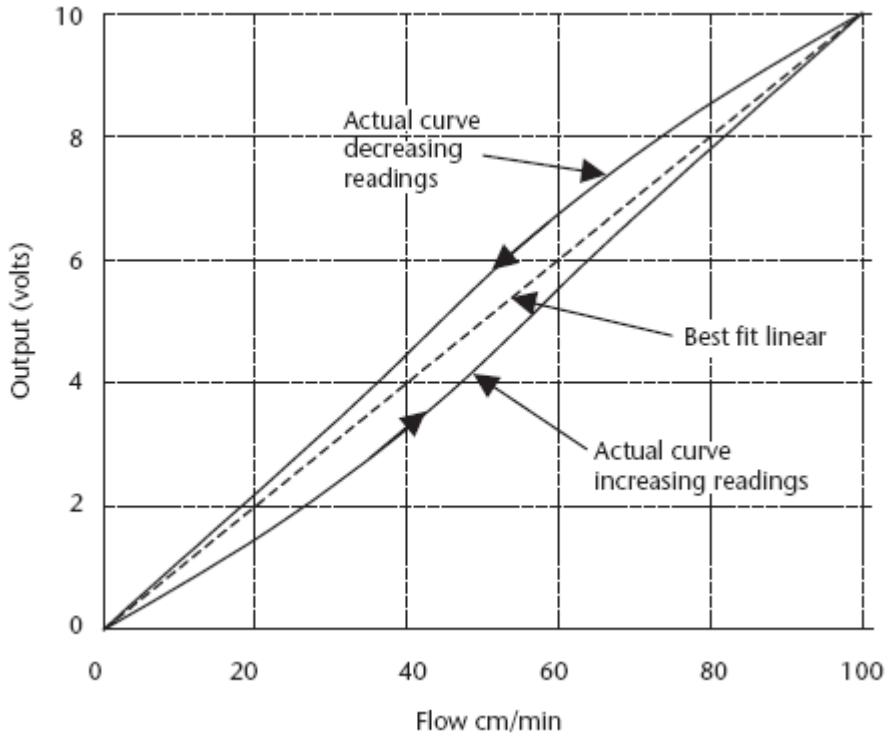


Figure 1.7:

Time constant of a sensor to a sudden change in a measured parameter falls into two categories, termed *first-order* and *second-order* responses.

The first-order response is the time the sensor takes to reach its final output after a transient change. For example, a temperature measuring device will not change immediately following a change in temperature, due to the thermal mass of the sensor and the thermal conductivity of the interface between the hot medium and the sensing element. The response time to a step change in temperature is an exponential given by:

$$A(t) = A_0 + (A_f - A_0)(1 - e^{-t/\tau})$$

where A(t) is the amplitude at time t, A_0 is the initial amplitude, A_f is the final amplitude, and τ is the time constant of the sensor.

Exercise 1.4

A linear pressure sensor has a time constant of 3.1 seconds, and a transfer function of 29 mV/kPa. What is the output after 1.3 seconds, if the pressure changes from 17 to 39 kPa? What is the pressure error at this time?

The second-order response occurs when the effect of a transient on the monitoring unit is to cause oscillations in the output signal before settling down. The response can be described by a second-order equation.

Other parameters used in instrumentation are *Range, Span, Precision, Offset, Drift, and Repeatability*. The definitions of these parameters are given at the end of the chapter.

1.5 Control System Evaluation

A general criterion for evaluating the performance of a process control system is difficult to establish. In order to obtain the quality of the performance of the controller, the following have to be answered:

1. Is the system stable?
2. How good is the steady state regulation?
3. How good is the transient regulation?
4. What is the error between the set point and the variable?

1.5.1 Stability

In a system that uses feedback, there is always the potential for stability. This is due to delays in the system and feedback loop, which causes the correction signal to be in-phase with the error signal change instead of out-of-phase. The error and correction signal then become additive, causing instability. This problem is normally corrected by careful tuning of the system and damping, but this unfortunately comes at the expense of a reduction in the response time of the system.

1.5.2 Regulation

The regulation of a variable is the deviation of the variable from the set point or the error signal. The regulation should be as tight as possible, and is expressed as a percentage of the set point. A small error is always present, since this is the signal that is amplified to drive the actuator to control the input variable, and hence controls the measured variable. The smaller the error, the higher the systems gain, which normally leads to system instability. As an example, the set point may be 120 psi, but the regulation may be 120 ± 2.5 psi, allowing the pressure to vary from 117.5 to 122.5 psi.

1.5.3 Transient Response

The transient response is the system's reaction time to a sudden change in a parameter, such as a sudden increase in material demand, causing a change in the measured variable or in the set point. The reaction can be specified as a dampened response or as a limited degree of overshoot of the measured variable, depending on the process, in order to return the measured variable to the set point in a specified time. The topic is covered in more detail in Process Control Chapter.

1.6 Analog and Digital Data

Variables are analog in nature, and before digital processing evolved, sensor signals were processed using analog circuits and techniques, which still exist in many processing facilities. Most modern systems now use digital techniques for signal processing.

1.6.1 Analog Data

Signal amplitudes are represented by voltage or current amplitudes in analog systems. Analog processing means that the data, such as signal linearization, from the sensor is conditioned and corrections that are made for temperature variations are all performed using analog circuits. Analog processing also controls the actuators and feedback loops. The most common current transmission range is 4 to 20 mA, where 0 mA is a fault indication.

Exercise 1.5
The pressure in a system has a range from 0 to 75 kPa. What is the current equivalent of 27 kPa, if the transducer output range is from 4 to 20 mA?

1.6.2 Digital Data

Signal amplitudes are represented by binary numbers in digital systems. Since variables are analog in nature, and the output from the sensor needs to be in a digital format, an analog to digital converter (ADC) must be used, or the sensor's output must be directly converted into a digital signal using switching techniques. Once digitized, the signal will be processed using digital techniques, which have many advantages over analog techniques, and few, if any, disadvantages. Some of the advantages of digital signals are: data storage, transmission of signals without loss of integrity, reduced power requirements, storage of set

points, control of multiple variables, and the flexibility and ease of program changes. The output of a digital system may have to be converted back into an analog format for actuator control, using either a digital to analog converter (DAC) or width modulation techniques.

1.6.3 Pneumatic Data

Pressure was used for data transmission before the use of electrical signals, and is still used in conditions where high electrical noise could affect electrical signals or in hazardous conditions where an electrical spark could cause an explosion or fire hazard. The most common range for pneumatic data transmission is 3 to 15 psi (20 to 100 kPa in SI units), where 0 psi is a fault condition.

1.6.4 Smart Sensors

The digital revolution also has brought about large changes in the methodology used in process control. The ability to cost-effectively integrate all the controller functions, along with ADCs and DACs, have produced a family of Smart Sensors that combine the sensor and control function into a single housing. This device reduces the load on the central processor and communicates to the central processor via a single serial bus (Fieldbus), reducing facility wiring requirements and making the concept of plug-and-play a reality when adding new sensors.

Definitions

Absolute Accuracy of an instrument is the deviation from true expressed as a number.

Accuracy of an instrument or device is the difference between the indicated value and the actual value.

Actuators are devices that control an input variable in response to a signal from a controller.

Automation is a system where most of the production process, movement, and inspection of materials are performed automatically by specialized testing equipment, without operator intervention.

Controlled or Measured Variable is the monitored output variable from a process, where the value of the monitored output parameter is normally held within tight given limits.

Controllers are devices that monitor signals from transducers and keep the process within specified limits by activating and controlling the necessary actuators, according to a predefined program.

Converters are devices that change the format of a signal without changing the energy form (e.g., from a voltage to a current signal).

Correction Signal is the signal that controls power to the actuator to set the level of the input variable.

Drift is the change in the reading of an instrument of a fixed variable with time.

Error Signal is the difference between the set point and the amplitude of the measured variable.

Feedback Loop is the signal path from the output back to the input, which is used to correct for any variation between the output level and the set level.

Hysteresis is the difference in readings obtained when an instrument approaches a signal from opposite directions.

Instrument is the name of any various device types for indicating or measuring physical quantities or conditions, performance, position, direction, and so forth.

Linearity is a measure of the proportionality between the actual value of a variable being measured and the output of the instrument over its operating range.

Manipulated Variable is the input variable or parameter to a process that is varied by a control signal from the processor to an actuator.

Offset is the reading of the instrument with zero input.

Precision is the limit within which a signal can be read, and may be somewhat subjective.

Range of an instrument is the lowest and highest readings that it can measure.

Reading Accuracy is the deviation from true at the point the reading is being taken, and is expressed as a percentage.

Repeatability is a measure of the closeness of agreement between a numbers of readings taken consecutively of a variable.

Reproducibility is the ability of an instrument to repeatedly read the same signal over time, and give the same output under the same conditions.

Resolution is the smallest change in a variable to which the instrument will respond.

Sensitivity is a measure of the change in the output of an instrument for a change in the measured variable.

Sensors are devices that can detect physical variables.

Set Point is the desired value of the output parameter or variable being monitored by a sensor; any deviation from this value will generate an error signal.

Span of an instrument is its range from the minimum to maximum scale value.

Transducers are devices that can change one form of energy into another.

Transmitters are devices that amplify and format signals, so that they are suitable for transmission over long distances with zero or minimal loss of information.